

VRET Biofeedback Pipeline

Real-time ECG / EDA stress-scoring server for acrophobia virtual-reality exposure therapy

Technical report — architecture, signal processing, and verified bug fixes

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Component	vret_server_v2_procedural_final.py (+ fake_opensignals.py dev stream)
Hardware	PLUX biosignalsplux — ECG + EDA, streamed over LSL (pyls)
Processing	NeuroKit2; 1000 Hz acquisition, 50 Hz control loop

1 What the pipeline does

The server reads two physiological signals — the electrocardiogram (ECG) and electrodermal activity (EDA) — from a PLUX biosignalsplux device while a participant undergoes virtual-reality exposure to heights. From these it derives a single, continuously updated **composite stress score**. That score is intended to drive the VR scene (in the acrophobia study, the altitude of a balloon): as the participant's physiological arousal rises above their own resting baseline, the exposure can be modulated in real time.

The design principle throughout is that every number is computed by a validated library call (NeuroKit2 for the biosignal math, pylsl for transport) rather than by hand-rolled signal processing. The script is deliberately procedural — it reads top to bottom with no custom signal-processing functions — so the data path is auditable at a glance.

2 Signals and derived measures

Measure	Library call	Meaning & role
HR	nk.ecg_clean → nk.ecg_peaks → nk.ecg_rate	Heart rate (BPM), mean over the HR window. Rises with arousal.
RMSSD	nk.ecg_clean → nk.ecg_peaks → nk.hrv_time["HRV_RMSSD"]	Short-term heart-rate variability (ms) over a 60 s moving window. Falls with stress (parasympathetic withdrawal).
EDA	EMA smoothing of the tonic level	Skin conductance (uS). Rises with sympathetic arousal / sweating.
S_t	weighted sum of the three deltas	Composite stress score: how far the participant is from their own baseline.

The composite score combines the three signals as deviations from the participant's frozen baseline. Each signal is first expressed as a z-score — its deviation divided by its own baseline spread — and the z-scores are then weighted:

$$S_t = 0.5 * z(\text{EDA}_{\text{phasic}}) + 0.3 * z(\text{HRV}) + 0.2 * z(\text{HR})$$

where $z(x) = (x - \text{baseline_mean_x}) / \text{baseline_sigma_x}$

EDA carries the largest weight because skin conductance is the most direct, fastest-responding index of sympathetic arousal; HRV and HR refine it. These weights are a deliberate study-design choice (EDA is the primary signal in acrophobia work), not a tunable default. Two important refinements (see §7) make the formula behave: the EDA term uses the *phasic* component of EDA, not the raw level, and every signal is z-scored against its own baseline spread before weighting so that no signal dominates merely because its raw numbers are larger.

The smoothed live S_t is compared against two thresholds placed on the participant's own resting S_t distribution, measured during baseline:

```
thresh_mild = mean_baseline + 1.28 * sigma_baseline (90th percentile)
thresh_high = mean_baseline + 2.33 * sigma_baseline (99th percentile)
```

The multipliers are genuine normal-distribution z-scores: 1.28 is the 90th percentile (top 10%) and 2.33 the 99th (top 1%). Two design points matter here. First, the thresholds are centred on the baseline *mean*, not on zero — resting S_t is generally not zero, so “1.28 sigma above zero” would flag a calm participant; “mean + 1.28 sigma” correctly means “beyond the 90th percentile of this person's rest.” Second, sigma is taken from the *raw* instantaneous S_t spread, not the smoothed series (see §4, Bug 7).

Three bands and the Unity command

Band	Condition	Meaning	Unity
calm	$S_t < \text{thresh_mild}$	at or near rest, coping	“increase”
stressed	$\text{thresh_mild} \leq S_t < \text{thresh_high}$	moderately aroused	“neutral”
ultra-stressed	$S_t \geq \text{thresh_high}$	strongly aroused / overwhelmed	“decrease”

The direction is the therapeutic-exposure convention: advance the stimulus (raise the balloon) when the participant is coping, hold when moderately aroused, and back off (lower it) when overwhelmed — so arousal never drives the feared stimulus further up. The command is currently computed and printed; transmission to Unity is the Stage 5 task.

3 Run-time architecture

The server runs as a single 50 Hz control loop that pulls sample chunks from the LSL inlet and proceeds through five phases:

Phase	What happens
1. Connect & map	Resolve the LSL stream; identify which physical channel is ECG and which is EDA by reading the channel labels from the stream metadata , with a corrected fixed-index fallback (digital=ch0, EDA=ch1, ECG=ch2).
2. Baseline (120 s)	Collect resting ECG and EDA into buffers; compute HR and RMSSD on sliding windows so their behaviour is visible during collection.
3. Sanity check	A passive, print-only beat-plausibility test confirms the channel mapped to ECG actually looks like ECG (see §4, Bug 5).
4. Freeze baseline	Compute and lock the reference statistics from the whole baseline: avg_hr, avg_hrv, avg_eda (with 3-sigma outlier cleaning), and the baseline S_t mean and spread from a sliding stress-score sweep. Print nk.ecg_quality, the valid-window count, and guard RMSSD plausibility.
5. Live (600 s)	Each tick: compute current HR / RMSSD / EDA, subtract the frozen baseline to get the three deltas, form S_t , and classify into the three bands. During the first ~60 s, before the live RMSSD window has filled, delta-HRV is held at 0 (HRV assumed at its baseline average) so the weights stay fixed and the score keeps the same scale — no faked HRV, no renormalisation, no dead first minute.

Window design

HR uses up to 30 s of recent ECG (a memory cap, not a smoothing target) and appears within a few seconds of beats being detected. RMSSD uses a genuine 60 s moving window: this matters because RMSSD is a root-mean-square of successive beat-to-beat differences, so a short window is statistically noisy (a 10 s window gives an estimate standard deviation around 56 ms; a 60 s window narrows that to roughly 12 ms). HR and RMSSD are recomputed every 25 ticks (about every 0.5 s).

Why HR and HRV are not available instantly

Unlike EDA — which is a level that can be read and smoothed sample-by-sample — HR and HRV are *derived from heartbeats*, and beats take time to accumulate. To put a number on heart rate at all, the detector must see several R-peaks; at a resting rate around 80 BPM that is a few seconds before the first HR value appears. HRV is far more demanding: RMSSD is a statistic *over a series of successive beat-to-beat differences*, so it is only meaningful once enough beats have been collected. The pipeline uses a 60 s window (~80–90 beats) because that is the span over which short-term RMSSD becomes a stable estimate rather than a noisy one. This is an inherent property of the measures, not a limitation of the implementation: there is simply no valid live RMSSD in the first minute, which is exactly why the live phase holds delta-HRV at 0 (its baseline assumption) until the window fills, rather than inventing a value or destabilising the score.

4 Defects found and fixed

The pipeline was inherited and rebuilt. Seven distinct defects were diagnosed, each verified empirically against real recordings before and after the fix. They are listed in the order the data path encounters them.

Bug 1 — EDA / ECG channel swap in the server

The original code hard-coded the channel indices the wrong way round, so the smooth EDA trace was fed to the R-peak detector and the ECG to the EDA path. The mapping is now resolved from the stream's channel **labels**, with a corrected fixed-index fallback. **Verified:** the swap produced an impossible RMSSD near 1038 ms; correct mapping gives HR 72, RMSSD 12 ms on the same data.

Bug 2 — ECG never cleaned before peak detection

Raw electrode ECG carries 50 Hz mains hum and baseline wander. `nk.ecg_peaks` expects a *cleaned* signal and finds zero peaks on the raw trace, which then crashes `nk.hrv_time` on an empty RR array. `nk.ecg_clean` (default 'neurokit' method: 0.5 Hz high-pass + powerline filter) is now applied before every peak detection, the canonical NeuroKit order. **Verified on the correct channel:** with cleaning, RMSSD 63 ms and maximum RR 964 ms (physiological); without it, RMSSD inflates to 106 ms and maximum RR balloons to 2110 ms from missed beats. Cleaning does real work and is kept.

Bug 3 — wrong `ecg_peaks` return value, and no plausibility guard

To mirror the NeuroKit repository examples exactly, all call sites now capture peaks, `info = nk.ecg_peaks(...)` and pass the first return (the signals DataFrame) to `hrv_time / ecg_rate`, rather than the `info` dict. This was verified numerically identical (RMSSD 12.719 either way) — a fidelity change, not a behavioural one. Separately, a physiological plausibility band was added: any RMSSD outside 5–300 ms is rejected as a detection failure and shown as “—” rather than reported as a measurement. `nk.ecg_quality` is printed at baseline so a weak ECG channel is visible.

Bug 4 — column swap baked into the dev stream (root cause)

The deepest defect was in the development broadcaster, `fake_opensignals.py`: it hard-coded EDA from one raw column and ECG from another *against the recording's own header*, which declares column 2 = ECG and column 3 = EDA. It therefore transmitted the ECG trace labelled “EDA” and vice-versa; the label-based server then fed EDA into the R-peak detector. The broadcaster now **parses the header JSON**, identifies each physical column by sensor name, applies the correct PLUX transfer function to each, and publishes on correctly-labelled LSL channels. **Verified directly on the file:** reading column 2 as ECG yields 589 beats, HR 93, RMSSD 63, RR 451–964 ms (real); reading column 3 as ECG yields RMSSD 1541 and RR 377–5770 ms (garbage).

Bug 5 — the sanity check itself was misfiring

After Bug 4 was fixed, a run produced correct physiology yet still printed a “mapping INVERTED” warning. The old check counted zero-crossings on the *raw* ECG, where 50 Hz mains (around 100 crossings/s) dominates and fools the comparison — it was measuring mains contamination, not signal type, while the real HR/RMSSD math runs on the *cleaned* ECG. The check was replaced with a **beat-plausibility** test: clean each channel, detect R-peaks, and measure the fraction of RR intervals within the human 300–1500 ms band (40–200 BPM). **Verified:** on the true ECG channel that fraction is 1.00 versus 0.67 on the EDA channel; the check is now silent on the correct mapping and still fires if the channels are genuinely swapped.

A zero-crossing high-pass variant was also tested as a discriminator and **rejected**: after filtering, both channels sit near 10–14 crossings/s and the ECG is marginally lower, so it would mislabel the correct orientation. Beat-plausibility is the reliable discriminator.

Bug 6 — thresholds measured from zero, not from baseline

The classifier originally compared S_t against $k \times \text{sigma_baseline}$, implicitly assuming resting S_t sits at zero. It does not: even at rest the weighted deltas average above zero (EDA in particular drifts upward slightly during baseline), so a calm participant was being flagged. The fix places the thresholds at **mean_baseline + $k \times \text{sigma_baseline}$** , which is what the z-score interpretation actually requires — “beyond the 90th/99th percentile of this participant’s own resting score.” **Verified:** after centring, a calm session opens in the calm band instead of immediately reading high.

Bug 7 — sigma collapsed by smoothing, pinning everything to ultra-stressed

A 14-minute test still read “ultra-stressed” almost throughout. The cause was that `sigma_baseline` was the standard deviation of the *smoothed* rolling-mean S_t series. Averaging shrinks variance (the standard deviation of a rolling mean is far smaller than that of the underlying values), so sigma came out near 0.4–0.9 when the true spread was several times larger. With sigma that small the two thresholds sat almost on top of each other and any real arousal leapt straight past “stressed” into “ultra-stressed.” The fix computes sigma from the *raw instantaneous* S_t values (their genuine spread) while still centring on the smoothed mean; both values are now printed for transparency. **Verified on the reference baseline:** raw sigma 6.22 versus smoothed 3.86, which widens the “stressed” band from ~4.1 to ~6.5 units and restores graded classification.

5 Verified output on the reference recording

All values below are from the real 384 s PLUX recording with the correct mapping and ECG cleaning in place.

Quantity	Value	Note
Heart rate	~88–93 BPM	Genuinely present in the recording, not an artifact.
RMSSD	~63–90 ms	Higher-normal at rest; fine as a baseline. VR exposure reads the deviation, not the absolute.
EDA (mean)	~3.6–6.6 uS	Plausible tonic skin-conductance range.
ECG quality	~0.83–0.85	nk.ecg_quality; good separation from poor-SNR ECG.
Baseline freeze	HR 88.2 / HRV 90.3 ms / EDA 3.60 uS	Thresholds use raw-s_inst sigma centred on the baseline mean (see Bug 7).

On the RMSSD “jumps”

A sudden RMSSD step such as 100 → 80 ms is normal, not a fault. RMSSD over a 60 s window (~93 beats) is a root-mean-square dominated by the few largest beat-to-beat differences; as the window slides every half second, a single large difference entering or leaving moves the value in visible steps. On this recording the sliding RMSSD ranged 34–101 ms with jumps up to ~12 ms between updates. Genuinely concerning patterns would instead be a value pinned constant (window not updating), values in the hundreds or thousands (detection failure, now guarded), or wild swings on every update (noisy detection). This is also why the composite score, not raw RMSSD, drives the exposure: it is deliberately smoothed.

6 Typical resting ranges for the three signals

For context when reading baseline values, the table gives broad population reference ranges for a healthy adult at rest. These are orientation figures, not pass/fail bounds: all three signals are strongly individual, so the pipeline always judges a participant against their own baseline rather than against population norms.

Signal	Typical resting range	Notes
Heart rate	~60–100 BPM	Conventional healthy-adult resting band; trained individuals are often lower.

Signal	Typical resting range	Notes
RMSSD (HRV)	~20–90 ms	Wide and age-dependent — commonly cited from below 20 to over 70 ms; declines with age. Highly individual.
EDA (tonic SCL)	~1–20 uS	Skin conductance level varies with hydration, temperature, and electrode placement; absolute value is far less informative than its change.

The reference recording (HR ~88 BPM, RMSSD ~90 ms, EDA ~3.6 uS at rest) sits comfortably inside these ranges.

7 EDA dominance: diagnosis and resolution

In extended testing the score spent most of the live phase in “ultra-stressed” even during quiet sitting. Investigation showed two compounding causes, both since addressed in code.

Cause 1 — raw EDA is dominated by tonic drift

EDA decomposes into a slow *tonic* level (SCL) and fast *phasic* responses (SCRs). On the reference recording, decomposition showed that roughly **99% of EDA variance was tonic** — the slow wander from electrode hydration and temperature, not event-driven arousal. Because the score compared the raw EDA level against a fixed baseline and weighted it at 0.5, this drift dominated S_t and pinned it high. **Resolution:** the pipeline now decomposes EDA with `nk.eda_phasic` on a rolling window and feeds only the *phasic* component into the score. After this change the EDA term contributes roughly 10% of the score instead of ~80%.

Cause 2 — signals on different scales

Even with phasic EDA, the three signals live on different numeric scales (phasic EDA in uS, HR and HRV in percent), with very different spreads, so the 0.5 / 0.3 / 0.2 weights did not act on comparable quantities.

Resolution: each signal is now z-scored against its own baseline spread before the weights are applied, so the weights express relative importance rather than an accident of scale. The professor’s weighting (EDA primary) is preserved; lowering the EDA weight was explicitly ruled out.

Remaining open item — validation on a truly calm recording

These fixes resolved EDA dominance, but testing then revealed that **HR and HRV** can themselves drive the score to “ultra-stressed” when the live phase differs physiologically from baseline (e.g. live RMSSD persistently ~40% below the baseline value). Because the baseline spreads can be small, z-scoring makes such sustained deviations register as very large. Whether this is correct behaviour (the score faithfully reporting genuine arousal) or over-sensitivity cannot be decided from the recordings collected so far, because each contained real physiological activity. The outstanding validation step is a recording in which baseline and live are *both genuinely calm and physiologically comparable*; if the live score then reads calm, the calibration is confirmed. If a sustained HRV drop persists while the participant is motionless, the cause is likely a baseline-versus-live setup difference (electrode settling, posture) to address at the protocol level rather than in the scoring math.

Other routes considered and not adopted: log-transforming EDA (dampens genuine responses too), an adaptive rolling EDA baseline (removes drift but also loses sustained elevation, a liability for exposure therapy), and capping the EDA term (arbitrary clip point). The two adopted fixes — phasic decomposition and per-signal z-scoring — are the principled, literature-aligned pair.

8 Notes for the move to the live device

- **Channel labels:** the live PLUX device provides no header file. Confirm its own LSL channel labels actually read “ECG” / “EDA”. If it exposes “CH1” / “CH2” instead, resolve ECG-vs-EDA by sensor-order logic rather than names, or the swap could reappear. The beat-plausibility sanity check is the safety net here.

- **Mains hum at source:** nk.ecg_clean removes 50 Hz, but a cleaner raw signal (good electrode contact, leads kept away from mains adapters) improves live-phase beat detection when the participant is moving.
- **EDA smoothing:** EDA_ALPHA = 0.005 at 1000 Hz is a fast (~ 0.2 s) EMA. Worth revisiting for responsiveness once results are stable.
- **Unity output (Stage 5, next):** the per-tick band command (“increase” / “neutral” / “decrease”) is currently computed and printed but not transmitted. The next step is to send it to the Unity VR scene — for a same-PC setup a local UDP socket is the simplest transport, with an LSL marker outlet as an alternative that matches the existing PLUX stack.
- **Calibration validation (open):** before the bands are trusted as absolute, collect one recording with a genuinely calm, physiologically comparable baseline and live phase (see §7) to confirm a calm participant reads calm.

Source of record: vret_server_v2_procedural_final.py and fake_opensignals.py. All quantitative claims in §4–§5 were reproduced in a sandbox against the reference recording.